Telematics Platform User Guide

5G NAD ConMod Module

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1 Introduction

This document provides usage information for Continental 5G NAD ConMod Module. The Continental 5G NAD ConMod Module provides a 5G (5th generation) cellular modem with a Linux based application processor. This document provides detailed usage information that is not included in the 5G NAD Module spec.

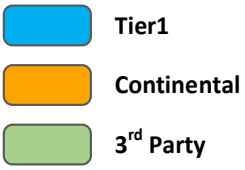
The target audience for this document is developers of components that interface with the NAD Module.

1.1 Notational Conventions

Ctrl+x	*A sequence with a '+' indicates that you must press and hold one key or pointing device button while pressing another key or pointing device button*
[ ]	*Optional elements in a command*
…..	*Indicates one of the following possibilities:* • *Additional optional arguments in a statement have been omitted* • *The preceding item or items can be repeated one or more times* • *Additional parameters, values, or other information can be entered*
-	*Line continuation: indicates that a command line or code line continues on the following line when at the end of a line*
Italic text	*Indicates important information, complete titles of manuals or variables*
`Mono-spaced`	*Example code or interactive screen display/output*
Numbers	*All numbers are assumed to be decimal unless otherwise noted. Non decimal radixes – binary, octal or hexadecimal – are explicitly indicated*
Underline blue text	*A link to an external reference (e.g., website, document)*

`Modules`
`|-> section`
`|-> item`
	A link to a section or item in the Telematics Platform html interface specification (Doxygen output from platform code headers)

Diagram Legend



2 Setup

2.1 Tools/Equipment Needed

- Continental 5G NAD ConMod Module
- 4-volt power supply (see 4V_PMX pin description in 5G NAD Module spec)
- Connection/access to NAD Module GPIO and peripherals of interest
- Continental Software Development Kit (SDK) – for on-target application development

2.2 Setup Instructions

The Continental 5G NAD ConMod Module should be physically connected to external peripherals (e.g., personal computer, microcontrollers, discrete components) to facilitate desired access to module functionality. At a minimum, an ability to control the 4-volt power supply and manipulate related power on/off signals (see Power Up/Down) should be supported. Access to other functionality such as UARTs and USB should be provided as applicable.

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2.3 Setup Troubleshooting

Not Applicable

3 Features/Components – Module Interface

This section details external interface to the 5G NAD ConMod module. The functionality described in this section would typically be provided by an external processor connected to the module, such as a vehicle bus controller. This is intended for developers working on an external controller for the 5G NAD ConMod module.

3.1 Software Update via fastboot

3.1.1 Overview

3.1.1.1 Platform Content Overview

The 5G NAD ConMod module supports a fastboot operation mode. This mode is used for flashing individual software packages from a Continental software release distribution. The fastboot executable is installed on Windows 10 and can access the NAD module when the module is in bootloader mode.

3.1.1.2 Product Expectations

- Install ADB (see https://developer.android.com/studio/command-line/adb) on Windows 10 Personal Computer (PC).
- Copy fastboot.exe from Continental release package to a local folder on Windows PC (recommend to include the folder in the path environment variable for easy access).
- Connect the NAD module HS USB port to the NAD module via a High Speed USB cable as specified in section **Fehler! Verweisquelle konnte nicht gefunden werden.**.

3.1.1.3 Extending Platform Functionality

Not Applicable

3.1.2 Setup

3.1.2.1 Tools/Equipment Needed

- Personal Computer (PC) with Windows 10 operating system
- winusb USB Host Driver (native on Windows 10)
- High Speed USB cable
- ADB (see https://developer.android.com/studio/command-line/adb)
- fastboot.exe (from Continental release package)

3.1.2.2 Setup Instructions

1. Install ADB following the instructions on https://developer.android.com/studio/command-line/adb on PC.
2. Copy fastboot.exe from Continental release package to local folder on PC.

3.1.2.3 Setup Troubleshooting

Not Applicable

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3.1.3 Usage

3.1.3.1 Basic Instructions

Basic update of Continental software release packages can be performed using the fastboot executable when the 5G NAD ConMod module is in bootloader mode. A Windows terminal (e.g., cmd window, Windows command line) is opened to access ADB and fastboot. The flashing sequence begins with using ADB to request the NAD module to enter bootloader mode. Once the NAD module is in bootloader mode, fastboot commands are used to flash individual software components. After the software components are updated, fastboot is used to trigger a reboot of the NAD module into normal operation mode. Persistence can then be updated using ADB when the NAD module is in normal operation mode.

3.1.3.2 Detailed Instructions

Note: Illustrated commands are performed within an open Windows terminal (e.g. cmd Window).

1 Put the 5G NAD ConMod module in bootloader mode

```
>adb reboot-bootloader
```

2 (Optional, Recommended) Verify the unit is in bootloader mode

```
>fastboot devices
```

3 (**One time only**) Update secure boot fuses to enable secure boot

```
>fastboot flash sec sec.dat
```

4 Update individual software components/partitions of the MTD and the UBI, or the complete UBI itself - Note that the component updates should be performed in the order specified. No other volumes than the ones listed below, should be flashed using fastboot.

```
                    MTD VOLUMES
>fastboot flash sbl <sbl file>.mbn
>fastboot flash tz_devcfg devcfg.mbn
>fastboot flash tz_devcfg_2 devcfg.mbn
>fastboot flash xbl_config xbl_cfg.elf
>fastboot flash xbl_config_2 xbl_cfg.elf
>fastboot flash multi_image multi_image.mbn
>fastboot flash multi_image_2 multi_image.mbn
>fastboot flash aop aop.mbn
>fastboot flash aop_2 aop.mbn
>fastboot flash qhee hyp.mbn
>fastboot flash qhee_2 hyp.mbn
>fastboot flash abl abl.elf
>fastboot flash abl_2 abl.elf
>fastboot flash uefi uefi.elf
>fastboot flash uefi_2 uefi.elf
>fastboot flash boot boot.img
> fastboot flash boot_2 boot.img
>fastboot flash tz tz.mbn
> fastboot flash tz_2 tz.mbn
                    UBI VOLUMES
```


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```
>fastboot flash system system.ubifs
>fastboot flash dsp2 dsp2.ubifs
                                    OR
>fastboot flash ubi ubi.img
```

NOTES:

- *<sbl file>* may be generic (e.g., sbl file is 'sbl1.mbn') or may be program specific (sbl file is 'sbl1-*myprogram*.mbn')
- Flashing the 'appl' component/partition (i.e., '*fastboot flash appl appl.ubifs*') is optional. This will overwrite any applications/data that are currently stored in this partition. This step should be skipped if 'appl' content needs to be maintained.
- If individual UBI volumes(system, dsp2, etc) are being flashed, there is no need to flash the complete 'ubi' partition with 'ubi.img'. Either of the steps mentioned above for UBI volumes are to be used.

5 Put the 5G NAD ConMod module in normal operation mode

```
>fastboot reboot
```

6 Update persistence if needed

```
>adb push <local folder>/tp-pers-install.tar.gz /tmp/tp-pers-install.tar.gz
>adb shell
[root@sa515m ]#pers-resource-installer /tmp/tp-pers-install.tar.gz
```

NOTES:

1 Partition table changes (e.g., repartitioning volumes, adding volumes) is not supported via fastboot. Software releases that include partition changes must be flashed in a different manner. Hence, the volume 'mibib' which generally contains partition related information should **NOT** be flashed using fastboot in releases that have partition related changes.
2 The modem Embedded File System (EFS) cannot be updated via fastboot (e.g., *.xqcn files).
3 The 'sec' partition contains fuses for secure boot. Once 'sec' is flashed the fuse settings cannot be undone or changed. The 'sec' partition update in step 3 should only be performed when secure boot is supported and needs to be enabled, since secure boot cannot be disabled. Also, RAM dumps cannot be obtained once secure boot is enabled.
4 Persistence must be updated the first time after secure boot is enabled via flashing the 'sec' partition (e.g., in same flashing sequence when 'sec' partition is flashed) due to persistence encryption.
5 For persistence update, the *<local folder>* is the folder on the PC where the persistence tarball `tp-pers-install.tar.gz` obtained from the release package is stored locally (e.g., "`D:\4K\`").

Example for flashing sequence that does not include 'sec' or 'appl' components, uses generic sbl file (`sbl1.mbn`), and does not include persistence update (prompts are **yellow**, input commands are **blue**, output/response is white):

```
>adb reboot-bootloader

>fastboot devices
6ee7c7b9        fastboot

>fastboot flash sbl sbl1.mbn
writing 'sbl'...
```


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```
OKAY [ 0.002s]
sending 'sbl' (521 KB)...
OKAY [ 0.132s]
finished. total time: 0.140s

>fastboot flash tz_devcfg devcfg.mbn
writing 'tz_devcfg'...
OKAY [ 0.003s]
sending 'tz_devcfg' (41 KB)...
OKAY [ 0.043s]
finished. total time: 0.051s

>fastboot flash xbl_config xbl_cfg.elf
writing 'xbl_config'...
OKAY [ 0.003s]
sending 'xbl_config' (38 KB)...
OKAY [ 0.035s]
finished. total time: 0.047s

>fastboot flash multi_image multi_image.mbn
writing 'multi_image'...
OKAY [ 0.003s]
sending 'multi_image' (12 KB)...
OKAY [ 0.042s]
finished. total time: 0.050s

>fastboot flash aop aop.mbn
writing 'aop'...
OKAY [ 0.002s]
sending 'aop' (136 KB)...
OKAY [ 0.049s]
finished. total time: 0.056s

>fastboot flash qhee hyp.mbn
writing 'qhee'...
OKAY [ 0.003s]
sending 'qhee' (82 KB)...
OKAY [ 0.047s]
finished. total time: 0.053s

>fastboot flash abl abl.elf
writing 'abl'...
```


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```
OKAY [ 0.003s]
sending 'abl' (148 KB)...
OKAY [ 0.050s]
finished. total time: 0.058s

>fastboot flash uefi uefi.elf
writing 'uefi'...
OKAY [ 0.003s]
sending 'uefi' (1292 KB)...
OKAY [ 0.245s]
finished. total time: 0.253s

>fastboot flash boot boot.img
writing 'boot'...
OKAY [ 0.002s]
sending 'boot' (7252 KB)...
OKAY [ 1.127s]
finished. total time: 1.134s

>fastboot flash tz tz.mbn
writing 'tz'...
OKAY [ 0.003s]
sending 'tz' (916 KB)...
OKAY [ 0.158s]
finished. total time: 0.166s

>fastboot flash ubi ubi.img
writing 'ubi'...
OKAY [ 0.004s]
sending 'ubi' (113408 KB)...
OKAY [ 17.869s]
finished. total time: 17.873s

>fastboot reboot
rebooting...

finished. total time: 0.007s
```

3.1.3.3 Testing/Error Reporting

Not Applicable.

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3.2 SSH terminal access lock/unlock

3.2.1 Overview

3.2.1.1 Platform Content Overview

The platform provides the possibility to lock SSH terminal access at runtime by blowing the End of line production security fuse using a provided utility.

SSH terminal access can be unlocked on commercial secure devices with the help of debug policy image.

3.2.1.2 Product Expectations

At the time of EOL OEMs must blow the production security fuse to prevent access to the target through SSH terminal.

This action is essential to ensure that production security cannot be bypassed.

Debug policy images can be used to enable SSH terminal access on production security enabled devices. `/.build/config/secbootcfg/debugpolicy/sdx55_debugpolicy.xml` file is used to configure, validate, and generate debug policy image (`apdp.mbn`), which enables debugging of images on target.

3.2.1.3 Extending Platform Functionality

Not Applicable

3.2.2 Setup

3.2.2.1 Tools/Equipment Needed

- Personal Computer (PC) with Windows 10 operating system
- winusb USB Host Driver (native on Windows 10)
- High Speed USB cable
- ADB (see https://developer.android.com/studio/command-line/adb)

3.2.2.2 Setup Instructions

Install ADB following the instructions on https://developer.android.com/studio/command-line/adb on PC.

3.2.2.3 Setup Troubleshooting

Not Applicable

3.2.3 Usage

3.2.3.1 Basic Instructions

Not Applicable

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3.2.3.2 Detailed Instructions

3.2.3.2.1 Lock SSH terminal access

To enable End of line production security on a device and lock SSH terminal access execute the following utility that is provided with a test image: */tst/bin/cas-enable-prod-security*
This will blow a corresponding production security hardware fuse. SSH terminal access will be locked on the next reboot cycle.

3.2.3.2.2 Unlock SSH terminal access

To unlock SSH terminal access on a production security enabled device, flash an appropriately configured debug policy image on to the target:

1 Update a debug policy configuration file */.build/config/secbootcfg/debugpolicy/sdx55_debugpolicy.xml* by:

a Set `DP_ENABLE_PROD_EOL_FUSE_OVERRIDE_OEM` flag to 1:

```
41          <flag> <bit_pos>60</bit_pos> <value>0</value> </flag> <!--DP_ENABLE_SUSPEND_SECUREBOOT_OEM-->
42          <flag> <bit_pos>61</bit_pos> <value>1</value> </flag> <!--DP_ENABLE_PROD_EOL_FUSE_OVERRIDE_OEM-->
43       </flags>
```

`DP_ENABLE_PROD_EOL_FUSE_OVERRIDE_OEM` flag:
· 0: (default): EOL fuse is not overridden
· 1: override production EOL fuse value to unlock SSH terminal access on a production security enabled device

b Add a device serial number need to be added to the list:

```
55        <serial_num_list> <!--Range: 1 to 200-->
56            <serial_num>0x2466e4ca</serial_num>
57        </serial_num_list>
```

It is possible to specify up to 200 unique serial numbers in this list field. All of the corresponding devices are then supported by the debug policy and subsystem ramdump will work.

1 Make a debug policy `apdp.mbn` image (see section “Suspend Secure Boot” of 5G NAD SecureBoot User Guide - TP_F101U-00050).

2 Copy signed debug policy image `apdp.mbn` from */release/images/devel/debugpolicy/sdx55/dbgp_ap/* to */release/images/devel/4K/*

3 Add following line to the file */release/images/devel/4K/rawprogram_nand_p4K_b256K.xml*:

```
<program PAGES_PER_BLOCK="64" SECTOR_SIZE_IN_BYTES="4096" filename="apdp.mbn"
label="apdp"          last_sector="7423"          num_partition_sectors="128"
physical_partition_number="0"                          start_sector="7296"/>
```

If the line is already in the file, modify it by setting `apdp.mbn` to the filename field.

4 Enable EDL mode on the target and flash the image from 4K folder using the supplier flashing tools.

3.2.3.3 Testing/Error Reporting

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3.3 Routing modem RAM dump to eMMC

3.3.1 Overview

3.3.1.1 Platform Content Overview

A RAM dump is produced when the modem crashes while NAD module application processor is running. This RAM dump captures contents of RAM memory associated with the modem.

This feature provides a possibility to route RAM dump files saving to eMMC storage. If the eMMC is populated, then eMMC partition of a specific name is created and formatted to be used as an associated mount point for `/data/ramdump` directory.

If the eMMC mount point doesn't exist due to missing eMMC device, the modem RAM dump will be stored on the NAD module application processor data partition under the same `/data/ramdump` path.

Note: Product should always populate the eMMC because /data volume can be low on size and because of size constraints RAM dumps cannot be always generated correctly and completely inside /data partition.

3.3.1.2 Product Expectations

The following expectations must be met in order to use the modem Ram dump routing to eMMC feature as designed:

- eMMC chip is populated on board, connected to external application processor
- minimum eMMC size requirement for the feature is 1Gb
- platform eMMC management (partitioning, formatting) is not modified without platform input/approval

Note: Only one RAM dump file is maintained in storage. New file collection due to a modem crash will overwrite previously stored file.

3.3.1.3 Extending Platform Functionality

Not Applicable

3.3.2 Setup

3.3.2.1 Tools/Equipment Needed

Not Applicable

3.3.2.2 Setup Instructions

Not Applicable

3.3.2.3 Setup Troubleshooting

Not Applicable

3.3.3 Usage

3.3.3.1 Basic Instructions

If the eMMC is present on the board, the eMMC will be partitioned and a RAM dump storage area will be formatted to VFAT. An eMMC primary partition will then mount automatically during each platform loading.

RAM dump files will be overwritten on every subsequent modem crash, so care must be taken to save the dump files from a previous crash before the next crash occurs.

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Note: Please refer to TP_F101U-00019 section 3.12 "Modem status" for more details on how modem reset status is reported by NADIF interface.

A typical partial RAM dump includes 2 files. For example,

```
[root@sa515m ramdump]#ls -l
ls -l
-rwxr-xr-x   1 root    root        4352 Oct  8 21:53 ramdump_microdump_modem.bin
-rwxr-xr-x   1 root    root   259228728 Oct  8 03:24 ramdump_modem.elf
```

ELF file is the main image for debugging purposes.

3.3.3.1.1 Generating a RAM dump file

Follow these steps:

1. Boot up the module
2. Trigger the modem crash by sending the following DIAG command:
   `sudo -u modem lte-test -c 123 75370300`
3. A modem RAM dump file is available in the device at the path: `/data/ramdump:`
   `ls -l /data/ramdump`

3.3.3.1.2 Confirm modem reset occurrence

Modem reset occurrence can be confirmed by looking at the module Kernel logs or debug logs.

Example kernel log (the module kernel log can be seen by typing "dmesg" in adb shell)

```
[root@sa515m ~]# [16033.060407] ipa-wan ipa3_del_dflt_wan_rt_tables:472 Deleting Route hdl:(0x9) with ip type: 0
[16033.061054] ipa-wan ipa3_del_dflt_wan_rt_tables:481 Deleting Route hdl:(0xb) with ip type: 1
[16033.068701] ipa-wan ipa3_wwan_remove:2955 rmnet_ipa completed deinitialization
[16033.077707] ipa ipa3_odl_pipe_cleanup:475 adpl pipe not configured
[16033.083434] ipa-wan ipa3_lcl_mdm_ssr_notifier_cb:3147 IPA BEFORE_SHUTDOWN handling is complete
[16033.089586] subsys-restart: subsystem_shutdown(): [kworker/u3:14:960]: Shutting down modem
[16033.115322] apr_tal_rpmsg qcom,glink:modem.apr_audio_svc.-1.-1: apr_tal_rpmsg_remove: Channel[apr_audio_svc] state[Down]
[16033.119262] [glink_pkt_poll]: DS channel in reset
[16033.127301] [glink_pkt_poll]: DATA40_CNTL channel in reset
[16033.130246] ipa-wan ipa3_lcl_mdm_ssr_notifier_cb:3150 IPA Received MPSS AFTER_SHUTDOWN
[16033.136663] ipa-wan ipa3_lcl_mdm_ssr_notifier_cb:3158 IPA AFTER_SHUTDOWN handling is complete
[16033.153560] [glink_pkt_read]: DATA40_CNTL channel in reset
[16033.187637] sysmon-qmi: ssctl_del_server: Connection lost between QMI handle and modem's SSCTL service
[16039.232857] ipa-wan ipa3_lcl_mdm_ssr_notifier_cb:3161 IPA received MPSS BEFORE_POWERUP
[16039.233572] ipa-wan ipa3_lcl_mdm_ssr_notifier_cb:3170 IPA BEFORE_POWERUP handling is complete
```


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[16039.239724] subsys-restart: subsystem_powerup(): [kworker/u3:14:960]: Powering up modem

[16039.251158] subsys-pil-tz 4080000.qcom,mss: modem: loading from 0x90800000 to 0xa0000000

[16039.710687] subsys-pil-tz 4080000.qcom,mss: modem: Brought out of reset

[16039.758791] subsys-pil-tz 4080000.qcom,mss: modem: Power/Clock ready interrupt received

[16039.767219] ipa-wan ipa3_lcl_mdm_ssr_notifier_cb:3173 IPA received MPSS AFTER_POWERUP

[16039.767509] rmnet_ipa3 started initialization

[16039.775194] IPA SSR support = True

[16039.778357] IPA is_platform_type_msm = False

[16039.782124] IPA SG support = False

[16039.786096] IPA Napi Enable = False

[16039.789301] using default for wan-rx-desc-size = 256

[16039.797211] ipa-wan ipa3_lcl_mdm_ssr_notifier_cb:3178 IPA AFTER_POWERUP handling is complete

[16039.798320] subsys-restart: subsystem_restart_wq_func(): [kworker/u3:14:960]: Restart sequence for modem completed.

[16039.823844] sysmon-qmi: ssctl_new_server: Connection established between QMI handle and modem's SSCTL service

[16039.859052] apr_tal_rpmsg qcom,glink:modem.apr_audio_svc.-1.-1: apr_tal_rpmsg_probe: Channel[apr_audio_svc] state[Up]

[16039.865785] diag: In diag_send_peripheral_buffering_mode, buffering flag not set for 0

[16040.108919] Sending QMI_IPA_INIT_MODEM_DRIVER_REQ_V01

[16040.136974] QMI_IPA_INIT_MODEM_DRIVER_REQ_V01 response received

[16040.137039] ipa ipa3_send_msg:544 wlan_msg_process failed

[16040.153415] ipa-wan ipa3_check_qmi_response:468 Got bad response 49 from request id 1 (error 48)

[16040.153454] ipa-wan rmnet_ipa_get_network_stats_and_update:3336 ipa3_qmi_get_network_stats failed: 1

Example debug log

Fatal error on modem!

[4947.617250] modem subsystem failure reason: err_inject_crash.c:411:Crash injected via Diag.

[4947.619519] subsys-restart: subsystem_restart_dev(): Restart sequence requested for modem, restart_level = RELATED.

[4947.628201] ipa-wan ipa3_lcl_mdm_ssr_notifier_cb:3129 IPA received MPSS BEFORE_SHUTDOWN

[4947.638174] ipa3_send_msg: 1 callbacks suppressed

[4947.638177] ipa ipa3_send_msg:544 wlan_msg_process failed

[4947.654601] ipa-wan ipa3_wwan_remove:2914 rmnet_ipa started deinitialization

[4947.690174] ipa-wan ipa3_wwan_remove:2929 rmnet_ipa unregister_netdev

[4947.690306] ipa-wan __ipa_wwan_close:1184 [rmnet_ipa0]: ipa3_deregister_intf failed -22

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```
[ 4947.760418] ipa-wan ipa3_del_dflt_wan_rt_tables:472 Deleting Route hdl:(0x9) with ip type: 0
[ 4947.761118] ipa-wan ipa3_del_dflt_wan_rt_tables:481 Deleting Route hdl:(0xb) with ip type: 1
[ 4947.768778] ipa-wan ipa3_wwan_remove:2955 rmnet_ipa completed deinitialization
[ 4947.777772] ipa ipa3_odl_pipe_cleanup:475 adpl pipe not configured
[ 4947.783545] ipa-wan ipa3_lcl_mdm_ssr_notifier_cb:3147 IPA BEFORE_SHUTDOWN handling is complete
[ 4947.789603] subsys-restart: subsystem_shutdown(): [kworker/u3:16:964]: Shutting down modem
[ 4947.817455] apr_tal_rpmsg qcom,glink:modem.apr_audio_svc.-1.-1: apr_tal_rpmsg_remove: Channel[apr_audio_svc] state[Down]
[ 4947.820652] [glink_pkt_poll]: DS channel in reset
[ 4947.828676] [glink_pkt_poll]: DATA40_CNTL channel in reset
[ 4947.832403] ipa-wan ipa3_lcl_mdm_ssr_notifier_cb:3150 IPA Received MPSS AFTER_SHUTDOWN
[ 4947.838464] ipa-wan ipa3_lcl_mdm_ssr_notifier_cb:3158 IPA AFTER_SHUTDOWN handling is complete
[ 4947.863430] [glink_pkt_read]: DATA40_CNTL channel in reset
[ 4947.899540] sysmon-qmi: ssctl_del_server: Connection lost between QMI handle and modem's SSCTL service
```

3.3.3.2 Detailed Instructions

Not Applicable

3.3.3.3 Testing/Error Reporting

If the feature is not working properly and modem RAM dump files are stored under NAD module application processor data partition, check the following:

- Check if eMMC device is present and working:
  `ls -l /dev | grep -i mmcblk0`
- Check if the eMMC primary partition is mounted under `/data/ramdump` path:
  `mount | grep -iw mmcblk0p1`

If the modem RAM dump file are not generated at all, check the following:

- if the Ram dump is enabled and a value of “1” is populated in the following file:
  `cat /sys/module/subsystem_restart/parameters/enable_ramdumps`
- if not execute the following command, reboot and try to test Ram dump once again:
  `echo 1 > /sys/module/subsystem_restart/parameters/enable_ramdumps`

- if the subsystem_ramdump process is running in background:
  `ps | grep -i subsystem_ramdump`
- if not run subsystem_ramdump manually and retest Ram dump
  `subsystem_ramdump 1 &`

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3.4 SSH terminal access in production

3.4.1 Overview

3.4.1.1 Platform Content Overview

The platform provides a possibility for SSH terminal access.

The root password login is solely controlled through production enable end of line fuse. In the production, this fuse would be blown and SSH service will not be run (see section 3.11).

3.4.1.2 Product Expectations

SSH server on a target is only run before EOL. SSH daemon is accessible when production security hardware fuse is not blown and test partition is present. At the time of EOL, a production security hardware fuse must be blown by OEM and the SSH server will not be available.

Debug policy images can be used to enable SSH terminal access on production security enabled devices after flashing the test partition (see section 3.11).

3.4.1.3 Extending Platform Functionality

Not Applicable

3.4.2 Setup

3.4.2.1 Tools/Equipment Needed

- Personal Computer (PC) with SSH Linux-Console

3.4.2.2 Setup Instructions

Connect to SSH server:1. Enable console using persistent key

```
persistent_key_access factory 53 integer CONSOLE-ENABLE
persistent_key_writer factory 53 CONSOLE-ENABLE integer 1
```

2. Connect to target via SSH:

```
ssh root@<ip address>
```

SSH logs into NAD soc linux console

3.4.2.3 Setup Troubleshooting

Not Applicable

3.4.3 Usage

3.4.3.1 Basic Instructions

Here's a generic command for a file transfer using SSH:

```
scp path/to/file root@<ip address>:/path/to/destination
```


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3.4.3.2 Detailed Instructions

3.4.3.2.1 Flashing the unit from the SSH Linux console

Telematics platform already integrates the MTD Utils which can be used for flashing the unit using SSH terminal access:

```
ubiformat /dev/mtd30 && ubiattach -p /dev/mtd30 && ubimkvol /dev/ubi1 -N appl -s
450MiB
mkdir /data/appfs; mount -t ubifs ubi1:appl /data/appfs/
```

push the images to the target using (/data/appfs/) scp

```
flash_eraseall /dev/mtd0
nandwrite -p /dev/mtd0 /data/appfs/sbl1.mbn
flash_eraseall /dev/mtd4
nandwrite -p /dev/mtd4 /data/appfs/apdp.mbn
flash_eraseall /dev/mtd8
nandwrite -p /dev/mtd8 /data/appfs/tz.mbn
flash_eraseall /dev/mtd12
nandwrite -p /dev/mtd8 /data/appfs/aop.mbn
flash_eraseall /dev/mtd14
nandwrite -p /dev/mtd8 /data/appfs/abl.elf
```

The UBI volumes can also be updated from the console.
First update the inactive bank/volumes, then switch to inactive bank/volumes and update initially active bank/volumes. The example below assumes primary volumes initially active, so secondary volumes updated first, then switch to secondary volumes and update primary volumes second::

```
ubirmvol /dev/ubi0 -N system_sec
ls -lrt /data/appfs/system_3.1.16.20.ubifs
-rw-rw-rw- 1 root root 72630272 Feb 26 2021 /data/appfs/system_3.1.16.20.ubifs
ubirmvol /dev/ubi0 -N uarea
ubimkvol /dev/ubi0 -s 72630272 -N system_sec
ubiupdatevol /dev/ubi0_6 /data/appfs/system_3.1.16.20.ubifs
```

CIFS shared drive can be also used for flashing:

```
mount -t cifs -o VERS=1.0,username=<username>,password=<password>,domain=<domain
name>,noserverino,nounix //<ip address>/<share name> <mount path>
```

Below are the logs of a successful attempt to flash from the mounted windows cifs shared drive:

```
flash_eraseall /dev/mtd0
flash_eraseall has been replaced by `flash_erase <mtddev> 0 0`; please use it
Erasing 256 Kibyte @ 240000 -- 100 % complete

nandwrite -p /dev/mtd0 /data/mnt1/Images/sbl1.mbn
```


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```
Writing data to block 0 at offset 0x0
Writing data to block 1 at offset 0x40000
Writing data to block 2 at offset 0x80000
```

Please ignore the message displayed with flash_eraseall - *flash_eraseall has been replaced by `flash_erase <mtddev> 0 0`; please use it*

3.4.3.2.2 Erasing test partition and enabling production security from the SSH Linux console

SSH terminal access can be used to execute a factory-reset script to blow the production security fuse and remove test partition at EOL:
/opt/conti/bin/factory-reset.sh

SSH server will not be run and terminal access will not be available on the next reboot cycle. It can be restored through debug policy configuration after flashing the test partition (see section 3.11).

3.4.3.3 Testing/Error Reporting

Not Applicable

3.5 Flashing support over ethernet

3.5.1 Overview

3.5.1.1 Platform Content Overview

Platform provides support to mount, share and flash images over ethernet based on Common Internet file system (CIFS) sharing. Once the remote files are mounted over SoC Linux Console, the same can be accessed and used for flashing.

This approach is not for flashing the blank NAD module or initial loading.

3.5.1.2 Product Expectations

The product team should set up the shared drive on desktop or another node which is accessible through ethernet link. The creation of shared drive on other side of ethernet node is not covered in this document.

For reference to create the shared drive in Ubuntu based desktop environment has been mentioned here - _Toc49370035

For reference to create Windows share folder or drive - _Toc49370035

The product team is also responsible for setting up the ethernet link. IP Address of the node over ethernet link should be accessible from the NAD console.

3.5.1.3 Extending Platform Functionality

Not Applicable

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3.5.2 Setup

3.5.2.1 Tools/Equipment Needed

- Personal Computer (PC) with Windows 10 operating system

3.5.2.2 Setup instructions

Not Applicable

3.5.2.3 Setup Troubleshooting

Not Applicable

3.5.3 Usage

3.5.3.1 Basic Instructions

Not Applicable

3.5.3.2 Detailed Instructions

Setup the ethernet link between NAD device and shared desktop or device where flashing images are kept.

1. Create a folder inside /data folder for mounting the share drive over network. Replace the mount point with any preferred path. Run this command on the NAD module.
   mkdir -p /data/<mount_point>

2. Mount the shared folder using cifs. Ip_addr should the ip address of the remote node/computer. Run this command on the NAD module
   mount -t cifs -o username=<user>,password=<pass> //<ip_addr>/<shared_folder_path> /data/<mount_point>

3. Once the mounting happens the images would be available on the Linux shell of the NAD target under /data/<mount_point>.
4. For flashing the images over the Linux shell or SSH, please refer the section - Flashing the unit from the SSH Linux console

3.5.3.3 Testing/Error Reporting

Not Applicable

3.6 Crash debug information in persistency for field return analysis

3.6.1 Overview

3.6.1.1 Platform Content Overview

The platform provides the possibility to save into a persistency storage a modem and linux kernel information facilitating debugging and fixing of problems, like a modem crash signature, timestamp, system kernel crash stack trace, and the amount of the modem crash in persistence. Moreover, some an auxiliary information like SWL history, flash scrubbing events, counter of EFS recovery events and temperature histogram are also stored in the persistency storage. Once a kernel/modem crash happens, the debug information will be available in the persistence after the reboot, and the device boot up completely. The debug information overrides once a new crash happens.

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3.6.1.2 Product Expectations

Persistency need to be installed for the Modem crash signature, modem crash timestamp, system kernel crash signature and system crashes counter to be stored in runtime:

1. Push persistency file to the board:
   adb wait-for-device push "tp-pers-install.tar.gz" /tmp/tp-pers-install.tar.gz
2. Install persistency:
   adb wait-for-device shell "pers-resource-installer /tmp/tp-pers-install.tar.gz
3. Reboot platform:
   adb wait-for-device shell "reboot"

3.6.1.3 Extending Platform Functionality

Not Applicable

3.6.2 Setup

3.6.2.1 Tools/Equipment Needed

- Personal Computer (PC) with Windows 10 operating system

3.6.2.2 Setup instructions

Not Applicable

3.6.2.3 Setup Troubleshooting

Not Applicable

3.6.3 Usage

3.6.3.1 Basic Instructions

Not Applicable

3.6.3.2 Detailed Instructions

In order to get the required data from the Persistency, a client need to check if the corresponding key exists. The **tpPCL_keyExist** function is used for that. If the key exists, so we can read its value from persistency by means of a function **tpPCL_keyReadData** (see the section “Persistence User Guide” – TP_F101U-00026).

3.6.3.2.1 Modem crash data

Modem crash signature and timestamp have the following keys in the Persistency:

NADIF_MODEM_CRASH_DATA – for modem crash signature

NADIF_MODEM_CRASH_TIMESTAMP – for timestamp of modem crash

Modem crash signature and timestamp example:

NADIF_MODEM_CRASH_TIMESTAMP

1546300938

NADIF_MODEM_CRASH_DATA

2019-01-01 00:02:17| modem subsystem failure reason: err_inject_crash.c:411:Crash injected via Diag.

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3.6.3.2.2 Kernel crash data

System kernel crash signature and count have the following keys in the Persistency:

KERNEL_DUMP_COUNT – for count of kernel crash signature

KERNEL_DUMP – for kernel crash signature

System kernel crash signature and count example:

KERNEL_DUMP

<4>[79.206897] CPU: 0 PID: 940 Comm: sh Tainted: G O 4.14.170LE.UM.4.1.1.c11-00200-sa515m-172-g8682345 #1

<4>[79.229129] Hardware name: Qualcomm Technologies, Inc. SDXPRAIRIE (Flattened Device Tree)

<4>[79.239285] task: 64bf0277 task.stack: 88a42e6e

<4>[79.247619] PC is at sysrq_handle_crash+0x1c/0x28

<4>[79.251952] LR is at 0x0

<4>[79.256810] pc : [<c051640c>] lr : [<00000000>] psr: 600b0013

<4>[79.259419] sp : e76cbe90 ip : 00000012 fp : 01f27548

<4>[79.265406] r10: 00000004 r9 : 00000002 r8 : 00000000

<4>[79.270616] r7 : 00000063 r6 : 0000000a r5 : c121aea0 r4 : c125ace8

<4>[79.275826] r3 : 00000001 r2 : 00000000 r1 : 00000000 r0 : 000f4240

<4>[79.282425] Flags: nZCv IRQs on FIQs on Mode SVC_32 ISA ARM Segment none

<4>[79.288935] Control: 10c53c7d Table: a9bbc059 DAC: 00000051

<0>[79.296137] Process sh (pid: 940, stack limit = 0xe702a704)

<0>[79.301866] Stack: (0xe76cbe90 to 0xe76cc000)

<0>[79.307249] be80: c125ace8 c0516928 00000002 01f29d18

<0>[79.311769] bea0: 00000001 00000000 01f29d18 00000002 00000004 c0516d70 c0516d0c eac91000

<0>[79.319928] bec0: 00000001 c029abe8 e68c0900 c029ab80 c1209088 e76cbf78 01f29d18 c0240fa4

<0>[79.328089] bee0: e76cbf78 01f29d18 00000002 00000004 e76cbf3c c0144e04 c0e89790 e68c0900

<0>[79.336248] bf00: 00000000 00ffffff 00000001 c0242de8 00000105 00000000 e76cbf78 00000000

<0>[79.344408] bf20: ebc9119c ebc9119c e76cbf78 85bdad46 00000002 e68c0900 00000002 00000000

<0>[79.352569] bf40: e76cbf78 01f29d18 00000002 c0241244 e68c0900 01f29d18 00000002 e68c0900

<0>[79.360727] bf60: 00000000 c1209088 e68c0900 01f29d18 00000002 c0241428 00000000 00000000

<0>[79.368886] bf80: 00000006 85bdad46 00000000 000b5bfc 00000000 00000002 00000004 c0107da4

<0>[79.377047] bfa0: e76ca000 c0107ba0 000b5bfc 00000000 00000001 01f29d18 00000002 00000000

<0>[79.385207] bfc0: 000b5bfc 00000000 00000002 00000004 00000001 01f29d18 00000001 01f27548

<0>[79.393366] bfe0: 00000078 be9f4848 0001b940 b6da921c 60070010 00000001 00000000 00000000

<4>[79.401535] [<c051640c>] (sysrq_handle_crash) from [<c0516928>] (__handle_sysrq+0x88/0x130)

<4>[79.409688] [<c0516928>] (__handle_sysrq) from [<c0516d70>] (write_sysrq_trigger+0x64/0x78)

<4>[79.417849] [<c0516d70>] (write_sysrq_trigger) from [<c029abe8>] (proc_reg_write+0x68/0x78)

<4>[79.426184] [<c029abe8>] (proc_reg_write) from [<c0240fa4>] (__vfs_write+0x28/0x140)

<4>[79.434515] [<c0240fa4>] (__vfs_write) from [<c0241244>] (vfs_write+0xb8/0x144)

<4>[79.442498] [<c0241244>] (vfs_write) from [<c0241428>] (SyS_write+0x5c/0xb8)

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<4>[79.449532] [<c0241428>] (SyS_write) from [<c0107ba0>] (ret_fast_syscall+0x0/0x54)
<0>[79.456821] Code: e59f2010 e5823000 f57ff04e e3a02000 (e5c23000)
KERNEL_DUMP_COUNT
1

TP_F101U_00026

3.6.3.2.3 SWL result history

The results of the latest ten SWL procedures are stored at the following persistency key:

SWL_HISTORY_LASTEST_TEN_ENTRIES – the SWL result history, latest 10 SWL updates

The result object consists of several fields:

update status – is the result of an update event, 1 byte, 0 - success
timestamp – is the timestamp of an update, 8 bytes, time in microseconds from 1 January of 1970
version – is the SWL version string, 33 bytes long, e.g **conmod-sa515m-3.3.255.0**

The result history array contains a queue of result objects. This means that the latest (think of it as the newest one) update result comes at the beginning of the history array. Therefore, the oldest one is stored at the end of the history array.

The number of all SWL events is stored in the following persistency key:

SWL_HISTORY_GENERAL_COUNTER – the number SWL events, 4 bytes, unsigned integer

The first SWL event is located in this key:

SWL_HISTORY_THE_FIRST_ENTRY – the first ever recorded SWL event result

The result object consists of several fields:

update status – is the result of an update event, 1 byte, 0 - success
timestamp – is the timestamp of an update, 8 bytes, time in microseconds from 1 January of 1970
version – is the SWL version string, 33 bytes long, e.g **conmod-sa515m-3.3.255.0**

3.6.3.2.4 Flash scrub timestamps

The flash scrubbing history is stored in the following persistency key:

LATEST_TEN_FLASH_SCRUBBING_TS – the array of last ten flash scrub results

The result type is an object consisting of two components:

timestamp – is the flash scrubbing event timestamp, microseconds from January 1, 1970, 8 bytes
event status – is the status of the flash scrubbing event, 4 bytes, 0 if flash scrubbing failed

The newest event is at the beginning of the array and the oldest one is at the end of the array.

The number of flash scrubbing events is stored in the following key:

FLASH_SCRUBBING_COUNT – is the number of the flash scrub events, is a 4 byte unsigned integer

The first flash scrubbing event is stored in the following key of persistency storage:

FIRST_FLASH_SCRUBBING_TS – is the timestamp of the first ever recorded flash scrub event

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The result type consists the following fields:

timestamp – is the timestamp of the first flash scrubbing event, microseconds from January 1, 1970, 8 bytes

event status – is the status of the first flash scrubbing event, 4 bytes, 0 if flash scrubbing failed

3.6.3.2.5 EFS recovery event

The number of EFS recovery events is stored in the following key of persistency storage:

EFS_RECOVERY_EVENT_COUNT – is the number of EFS recovery events, 4 bytes

The EFS recovery failure is stored in the following key:

EFS_RECOVERY_FAILURE – indicates that EFS recovery procedure failed

The value consists of the following fields:

failure status – is the EFS recovery failure status, 4 bytes

timestamp – is the EFS recovery timestamp, 8 bytes, microseconds from the 1 Jan 1970

3.6.3.2.6 Temperature histogram

The temperature histogram range values are stored in the following keys

Note: each value is a 4 byte integer

TEMPERATURE_RANGE_0 – value for t < -40^{0}C
TEMPERATURE_RANGE_1 – value for -40^{0}C < t < -30^{0}C
TEMPERATURE_RANGE_2 – value for -30^{0}C < t < -20^{0}C
TEMPERATURE_RANGE_3 – value for -20^{0}C < t < -10^{0}C
TEMPERATURE_RANGE_4 – value for -10^{0}C < t < 0^{0}C
TEMPERATURE_RANGE_5 – value for 0^{0}C < t < 10^{0}C
TEMPERATURE_RANGE_6 – value for 10^{0}C < t < 20^{0}C
TEMPERATURE_RANGE_7 – value for 20^{0}C < t < 30^{0}C
TEMPERATURE_RANGE_8 – value for 30^{0}C < t < 40^{0}C
TEMPERATURE_RANGE_9 – value for 40^{0}C < t < 50^{0}C
TEMPERATURE_RANGE_10 – value for 50^{0}C < t < 60^{0}C
TEMPERATURE_RANGE_11 – value for 60^{0}C < t < 70^{0}C
TEMPERATURE_RANGE_12 – value for 70^{0}C < t < 80^{0}C
TEMPERATURE_RANGE_13 – value for 80^{0}C < t < 90^{0}C
TEMPERATURE_RANGE_14 – value for 90^{0}C < t < 100^{0}C
TEMPERATURE_RANGE_15 – value for 100^{0}C < t < 102^{0}C
TEMPERATURE_RANGE_16 – value for 102^{0}C < t < 104^{0}C
TEMPERATURE_RANGE_17 – value for 104^{0}C < t < 106^{0}C
TEMPERATURE_RANGE_18 – value for 106^{0}C < t < 108^{0}C
TEMPERATURE_RANGE_19 – value for 108^{0}C < t < 110^{0}C
TEMPERATURE_RANGE_20 – value for 110^{0}C < t < 112^{0}C
TEMPERATURE_RANGE_21 – value for 112^{0}C < t < 114^{0}C

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TEMPERATURE_RANGE_22 – value for 114^{0}C < t < 116^{0}C
TEMPERATURE_RANGE_23 – value for 116^{0}C < t < 118^{0}C
TEMPERATURE_RANGE_24 – value for 118^{0}C < t < 120^{0}C
TEMPERATURE_RANGE_25 – value for t > 120^{0}C

3.6.3.3 Testing/Error Reporting

Not Applicable

4 Known Issues

Not Applicable

5 FAQs

Not Applicable

6 Platform DocumentationTroubleshooting/Tips and Tricks

Not Applicable

7 References

7.1 Abbreviations/Acronyms/Definitions

Not Applicable

7.2 Platform Documentation

TP_102G-00001: Telematics Platform Glossary
TP_F101U-00019: User Guide NADIF
TP_F101U-00023: Common Service User Guide
TP_F101U-00026: Persistence User Guide
TP_F101U-00035: Software Development Kit User Guide
TP_F101U-00039: 5G NAD Tier 1 Power Management User Guide
TP_F101U-00050: 5G NAD SecureBoot User Guide
TP_F103U-00034: User Guide 5G NAD Module
5G NAD Module spec: TP_LTE_5G_SA515M_NAD_Specification

7.3 Website links

- Android Debug Bridge (ADB): https://developer.android.com/studio/command-line/adb
- Common Internet File System (CIFS) : https://cifs.com/
- Setting up Samba Share on Ubuntu : https://ubuntu.com/tutorials/install-and-configure-samba#1-overview
- Setting up windows 10 share : https://support.microsoft.com/en-us/windows/file-sharing-over-a-network-in-windows-10-b58704b2-f53a-4b82-7bc1-80f9994725bf

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8 Revision History

Version	Date	Description
01.00	Sep 27, 2021	Initial release
02.00	Jun 28, 2022	Modem crash and kernel crash debug information in persistency
03.00	Dec 23, 2022	SWL, EFS, Flash scrubbing and temperature histogram information in persistency

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